

TEST CODE **01238010**

JANUARY 2006

FORMTP2006021**CARIBBEAN EXAMINATIONS COUNCIL****SECONDARY EDUCATION CERTIFICATE
EXAMINATION****PHYSICS****Paper01 – General Proficiency***75 minutes***17 JANUARY 2006 (a.m.)****READ THE FOLLOWING DIRECTIONS CAREFULLY**

1. In addition to this test booklet, you should have an answer sheet.
2. Each item in this test has four suggested answers lettered (A), (B), (C), (D). Read each item you are about to answer and decide which choice is best.
3. On your answer sheet, find the number which corresponds to your item and shade the space having the same letter as the answer you have chosen. Look at the sample item below.

Sample Item

The SI unit of length is the

- (A) newton
(B) metre
(C) kilogram
(D) second

Sample Answer

- (A) ☒ (B) ☐ (C) ☐ (D) ☐

The best answer to this item is "metre" so answer space (B) has been shaded.

4. If you want to change your answer, be sure to erase your old answer completely and fill in your new choice.
5. When you are told to begin, turn the page and work as quickly and as carefully as you can. If you cannot answer an item, omit it and go on to the next one. You can come back to the harder item later. Your score will be the total number of correct answers.
6. Figures are not necessarily drawn to scale.
7. You may do any rough work in this booklet.
8. The use of non-programmable calculators is allowed.
9. This test consists of 60 items. You will have 75 minutes to answer them.
10. Do not be concerned that the answer sheet provides spaces for more answers than there are items in this test.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

↑ AFFIX SEAL HERE ↓

- (A) 0.0634
 (B) 0.00634
 (C) 0.000634
 (D) 0.0000634

2. Which of the following units is used to measure kinetic energy?

- (A) The newton (N)
 (B) The watt (W)
 (C) The joule (J)
 (D) The pascal (P)

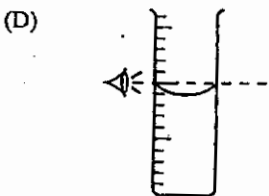
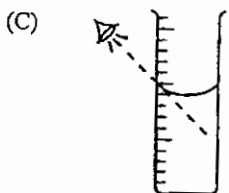
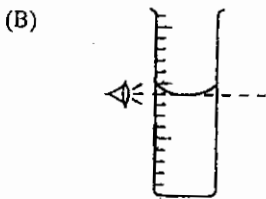
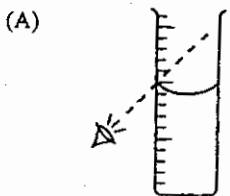
3. Which of the following is NOT a fundamental physical quantity?

- (A) Mass
 (B) Time
 (C) Force
 (D) Length

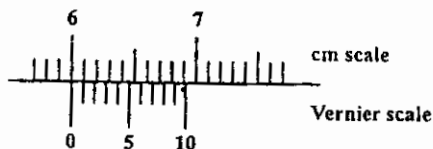
4. Which of the following has NO units?

- (A) Velocity
 (B) Relative density
 (C) Energy
 (D) Momentum

5. Which of the following diagrams illustrates the correct method of reading the liquid level?



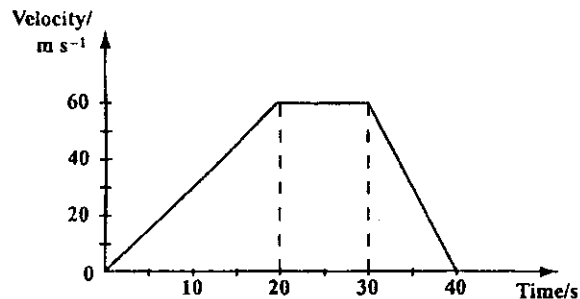
6.



The diagram above indicates a vernier caliper measurement of

- (A) 6.00 cm
 (B) 6.02 cm
 (C) 6.10 cm
 (D) 6.11 cm

7. The following graph shows the journey made by a motorist. During which time interval is his velocity constant?



- (A) Between 0 and 20 secs
(B) Between 20 and 30 secs
(C) Between 30 and 40 secs
(D) Between 0 and 40 secs

8. In which of the following is the quantity correctly paired with its unit?

Quantity

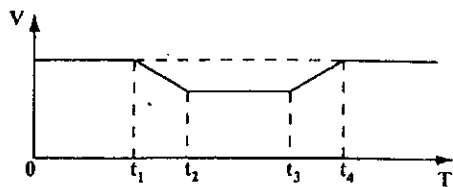
Unit

- | | |
|---------------------|------------------|
| (A) Mass | Newton |
| (B) Moment of force | Newton per metre |
| (C) Energy | Newton-metre |
| (D) Work done | Newton per metre |

9. A single force, which can replace and have the same effect as two or more forces acting together, is known as the

- (A) resultant
(B) equilibrant
(C) moment
(D) vector

10.



The diagram shows a velocity/time graph for a moving object. Which of the following statements about the object is/are true?

- I. It returns to its starting point.
II. It has zero acceleration between times t_2 and t_3 .
III. Its velocity at t_4 is the same as its initial velocity.

- (A) II only
(B) I and II only
(C) II and III only
(D) I, II and III

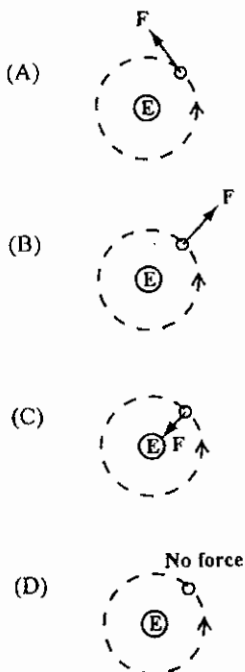
11. Which of the following could be used to find the speed of an object?

- (A) $\frac{\text{Change in velocity}}{\text{Time taken}}$
(B) $\frac{\text{Change in displacement}}{\text{Time taken}}$
(C) $\frac{\text{Distance travelled}}{\text{Time taken}}$
(D) Distance travelled x Time taken

12. A falling raindrop reaches a constant speed when

- (A) there is no net force acting on it
(B) the pull of the earth on the raindrop is equal to the weight of the raindrop
(C) the upthrust due to the air is at a minimum
(D) the air surrounding the raindrop becomes saturated with water vapour

13. A communications satellite is orbiting Earth, E. Which of the following diagrams correctly shows the resultant force, F , on it?



14. When two bodies collide, momentum is conserved. This means that the
- kinetic energy before impact is equal to that after impact
 - momentum of each body is unchanged
 - algebraic sum of the velocities before impact is equal to the sum of the velocities after impact
 - total momentum of the bodies before impact is equal to the total momentum of the bodies after impact

15. For a swinging pendulum, which of the following is true when it is at its lowest point during an oscillation?

	Potential energy	Kinetic energy
(A)	Minimum	Maximum
(B)	Minimum	Minimum
(C)	Maximum	Maximum
(D)	Maximum	Minimum

16. Which of the following implements is/are designed to take advantage of a large moment provided by a relatively small force?

- Claw hammer
 - Crowbar
 - Pair of tweezers
 - Pair of wire cutters
- III only
 - I and IV only
 - I, II and III only
 - I, II and IV only

17. The rate of which of the following determines the power of a machine?

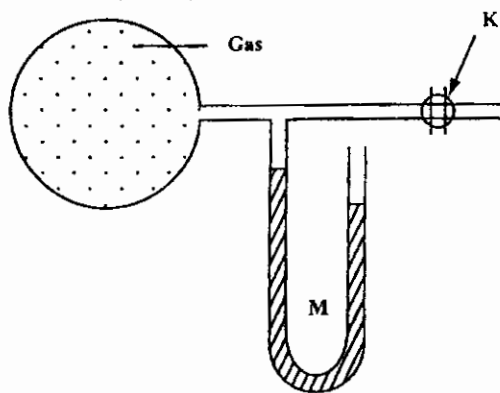
- Doing work
 - Converting energy
 - Changing temperature
- I only
 - III only
 - I and II only
 - I, II and III

18. A solid object, X, floats in mercury and sinks in water. A solid object, Y, floats in both mercury and water.

Which of the following is true about X and Y?

- X is less dense than Y.
- X and Y are both denser than water.
- X and Y are both denser than mercury.
- X is denser than Y.

19. The globe shown in the diagram below contains a gas whose pressure is being measured with a manometer, M.



Which of the following statements is/are correct?

- I. The pressure of the gas is less than that of the atmosphere.
- II. For a small pressure difference, mercury may be used in the manometer for an accurate reading of the gas pressure.
- III. In order to bring the two menisci in the manometer to the same level one could open the stop-cock, K.

- (A) I only
- (B) II only
- (C) I and II only
- (D) I and III only

20. A block is allowed to fall freely towards the ground. As it falls, its gravitational potential energy

- (A) remains constant
- (B) increases
- (C) is converted to kinetic energy
- (D) is converted to internal energy

21. To check the upper fixed point on the Celsius scale, a thermometer is placed in

- (A) boiling water at atmospheric pressure
- (B) melting ice at atmospheric pressure
- (C) steam above boiling water at atmospheric pressure
- (D) water in contact with melting ice at atmospheric pressure

22. A fixed mass of gas is compressed at constant temperature. The molecules of the gas will

- (A) hit the sides of the container more often
- (B) have more kinetic energy
- (C) move more slowly
- (D) have more momentum

23. Charles' law describes the variation of

- (A) volume with pressure of a fixed mass of gas at a constant temperature
- (B) volume with mass of a gas at constant temperature and pressure
- (C) pressure with temperature of a fixed mass of gas at constant volume
- (D) volume with temperature of a fixed mass of gas at constant pressure

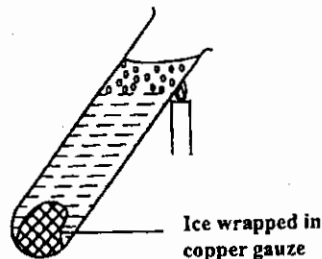
24. The kinetic theory of matter suggests that the molecules of a gas are

- (A) closely packed and vibrating
- (B) closely packed and moving over each other
- (C) far apart and vibrating about a fixed point
- (D) far apart and moving at random

25. A student wishes to use solar energy to boil an egg in the shortest possible time by placing a pot containing the egg at the focal point of a concave reflector. Which of the following combinations of surfaces should the student choose?

(A) A shiny white pot and a dull black reflector
 (B) A dull black pot and a shiny white reflector
 (C) A dull white pot and a shiny black reflector
 (D) A shiny black pot and a dull white reflector

26.



The figure above shows a piece of ice wrapped in copper gauze and submerged in a glass test tube of water. The water when heated at the top boils long before the ice is melted. The MAIN reason for this is that

(A) the test tube is a poor conductor of heat
 (B) copper is a good conductor of heat
 (C) ice is less dense than water
 (D) water is a poor conductor of heat

27. Which of the following are reasons why a hot liquid, placed in a double-walled vacuum flask, retains its heat for a long time?

I. Evacuated space between the double walls reduces the loss of heat by conduction.
 II. Silver inner walls reduce the loss of heat by radiation.
 III. The silvered outer wall helps to absorb heat from the surroundings.

(A) I and II only
 (B) I and III only
 (C) II and III only
 (D) I, II and III

28. Solar heating panels use the sun's radiation to warm water. The panels are blackened because dark surfaces are good

(A) reflectors of heat
 (B) transmitters of heat
 (C) absorbers of heat
 (D) conductors of heat

29. Which of the following would be the basic difference between a transverse mechanical wave and a longitudinal mechanical wave?

I. Medium through which they travel
 II. Direction of vibration of particles in medium
 III. Wavelength

(A) I only
 (B) II only
 (C) I and II only
 (D) II and III only

30. The spreading of a wave into the region behind an obstruction is called

(A) diffraction
 (B) refraction
 (C) superposition
 (D) interference

31. Which of the following will change when a wave is refracted at a boundary?

I. Velocity
II. Wavelength
III. Frequency

- (A) I only
(B) III only
(C) I and II only
(D) I, II and III

32. Sound waves travel fastest in

- (A) steel
(B) water
(C) air
(D) a vacuum

33. Which row in the table below correctly compares X-rays and visible light?

	X-rays	Visible Light
(A)	Particles	Waves
(B)	High frequency	Low frequency
(C)	Long wavelength	Short wavelength
(D)	Carry less energy	Carry more energy

34. The table below lists the refractive indices of four different materials.

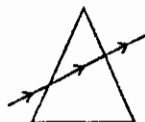
Material	Refractive Index
Air	1.0
Ice	1.3
Perspex	1.5
Diamond	2.4

In which medium would the light waves have the slowest speed?

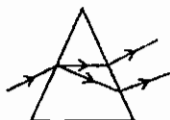
- (A) Air
(B) Ice
(C) Perspex
(D) Diamond

35. Which of the diagrams BEST represents the passage of a beam of white light through a triangular glass prism?

(A)



(B)



(C)



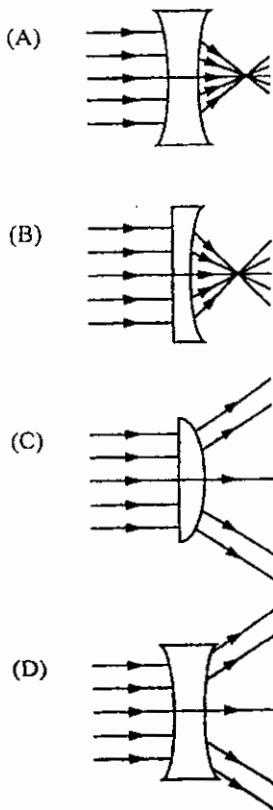
(D)



36. A ray of light in air strikes a glass block at an angle of incidence of 0° . The light will be

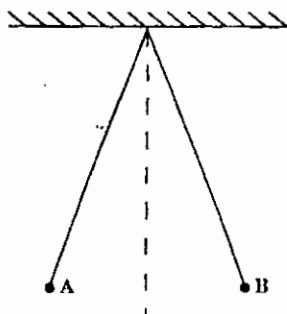
- (A) undeviated
(B) totally reflected
(C) refracted at 90° to normal
(D) refracted at an unknown angle

37. Which of the following diagrams BEST shows the behaviour of light rays passing through a glass lens in air?



38. If a converging lens is used as a magnifying glass, the image formed is
- (A) real, inverted, diminished
(B) real, erect, diminished
(C) virtual, inverted, magnified
(D) virtual, erect, magnified

39.



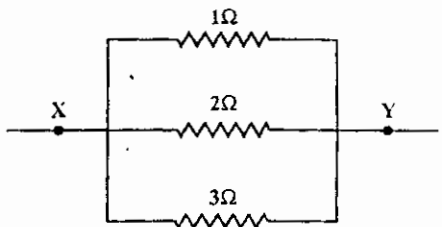
Two light, equally charged aluminium spheres A and B are suspended by insulating threads. If they come to rest as shown in the diagram, the force keeping them apart is

- (A) gravitational force
(B) electrostatic force
(C) magnetic force
(D) centripetal force
40. Which of the following ratios could represent the ampere?
- I. $\frac{\text{Coulomb}}{\text{Second}}$
II. $\frac{\text{Volt}}{\text{Coulomb}}$
III. $\frac{\text{Ohm}}{\text{Seconds}}$
- (A) I only
(B) I and III only
(C) II and III only
(D) I, II and III

41. The 'glasshouse' effect refers to the glasshouse acting as a heat trap. This is a direct result of

- (A) short wavelength infrared rays which are unable to penetrate glass
- (B) long wavelength infrared rays which are unable to penetrate glass radiated from the sun
- (C) short wavelength infrared rays which are unable to penetrate glass radiated from the objects within the glasshouse
- (D) long wavelength infrared rays which are unable to penetrate glass radiated from the objects within the glasshouse

44.



Which of the following statements for the parallel circuit in the diagram is/are true?

- I. The potential difference across each of the resistors is the same.
- II. The potential difference between X and Y is the same as that across the 2Ω resistor.
- III. The current through each of the resistors is different.

42. The resistance of an IDEAL ammeter is assumed to be

- (A) zero
- (B) 0.1 ohm
- (C) 1 to 2 ohm
- (D) infinite

- (A) I only
- (B) II only
- (C) I and III only
- (D) I, II and III

43. Which of the following formulae (in which the symbols have their usual meaning) is a mathematical expression of Ohm's Law?

- (A) $V = IR$
- (B) $P = VI$
- (C) $Q = IT$
- (D) $E = PT$

45. Which of the following circuit symbols represents a fuse?

- (A)
- (B)
- (C)
- (D)

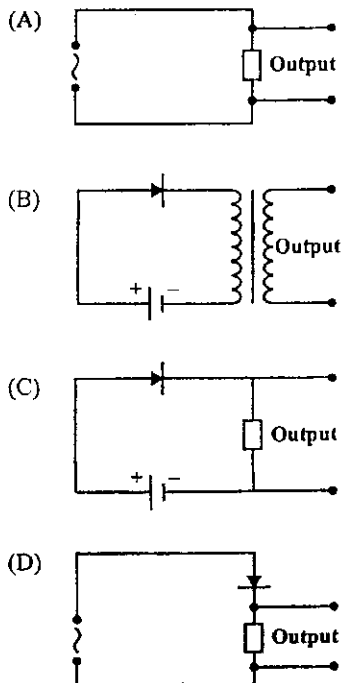
46. An appliance is connected to the mains by a cable which has three wires. What colour should the insulation be on the lead connected to the fuse?

- (A) Brown
- (B) Black
- (C) Blue
- (D) Green/yellow

47. Rectification can BEST be done by using a

- (A) transformer
- (B) capacitor
- (C) transistor
- (D) diode

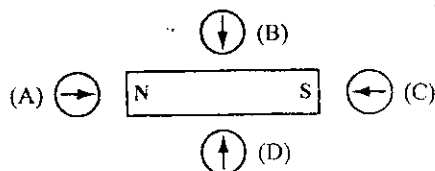
48. Which of the following circuits can be used for half-wave rectification?



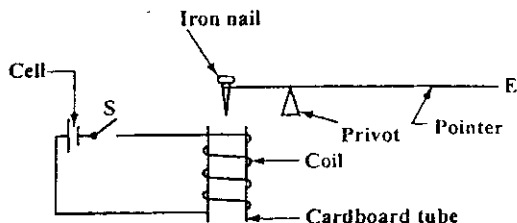
49. When two magnets, touching at their N and S poles, are separated, the force between them

- (A) increases
- (B) remains the same
- (C) decreases
- (D) first increases and then decreases

50. Ignoring the earth's magnetism, which of (A), (B), (C) or (D), correctly shows the direction in which a plotting compass would point when placed near the bar magnet shown below?



51.



In the apparatus shown in the diagram above, when the switch, S, is closed, the nail is pulled into the coil and the other end, E, of the pointer rises. This device uses which of the following effects?

- I. Motor effect
- II. Electromagnetic induction
- III. Electrostatic induction
- IV. Electromagnetism

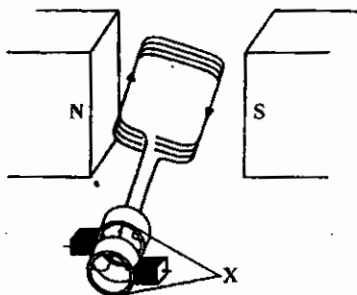
- (A) I and II only
- (B) II and IV only
- (C) I, II and III only
- (D) II, III and IV only

52. Which of the following would induce an e.m.f. in a conductor?

- I. Moving a magnetic field across the conductor
- II. Moving the conductor in the magnetic field of a magnet
- III. Placing the conductor near to a wire carrying an alternating current

- (A) I only
(B) I and II only
(C) I and III only
(D) I, II and III

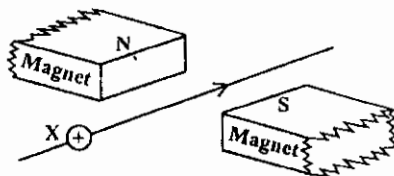
53.



Which of the following does the diagram above represent?

- (A) An a.c. generator
(B) A d.c. generator
(C) A voltmeter
(D) An ammeter

54.



A positively charged particle, X, is moving at high speed in the direction indicated. In which direction will the magnetic field deflect the particle?

- (A) Upward
(B) To the left
(C) Downward
(D) To the right

55. In which of the following devices is a commutator used?

- (A) Transformers
(B) a.c. generators
(C) d.c. motors
(D) Moving coil microphones

56. Which of the following scientists discovered that the nucleus also contained neutrons?

- (A) Niels Bohr
(B) James Chadwick
(C) Marie Curie
(D) Joseph Thompson

57. The nuclide ${}_{90}^{231}\text{Th}$ contains

- (A) 90 protons and 234 neutrons
(B) 235 protons and 90 neutrons
(C) 90 protons and 144 neutrons
(D) 144 protons and 90 neutrons

58. When alpha particles, beta particles and gamma rays from a radioactive source are subjected to a transverse magnetic field, which of them would NOT be deviated from their original path?
- I. Alpha
II. Beta
III. Gamma
- (A) I only
(B) II only
(C) I and II only
(D) II and III only
59. Which of the following equations for nuclear reactions is correct?
- (A) ${}^{226}_{88}\text{Ra} \rightarrow {}^{222}_{86}\text{Rn} + \beta\text{-particle}$
(B) ${}^{14}_6\text{C} \rightarrow {}^{15}_7\text{N} + \beta\text{-particle}$
(C) ${}^{226}_{88}\text{Ra} \rightarrow {}^{222}_{86}\text{Rn} + \alpha\text{-particle}$
(D) ${}^{14}_6\text{C} \rightarrow {}^{15}_7\text{N} + \alpha\text{-particle}$
60. Which of the following scientists discovered the relationship $E=mc^2$?
- (A) Marie Curie
(B) Isaac Newton
(C) Albert Einstein
(D) J. J. Thompson

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.